

# Machine Learning for Healthcare

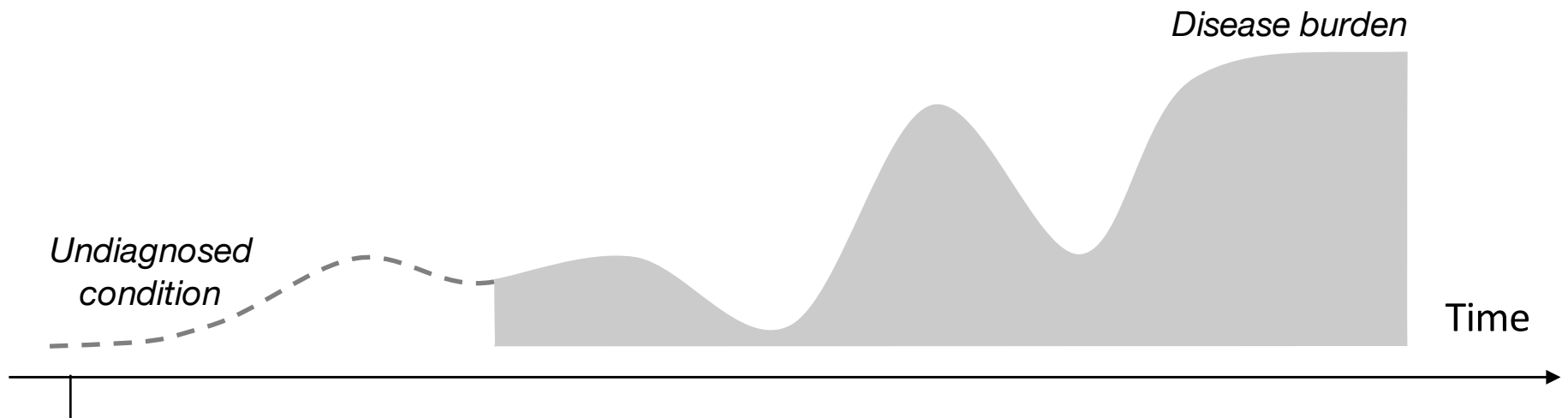
HST.956, 6.S897

## Lecture 18: Disease progression modeling & subtyping, Part 1

David Sontag



**Prognosis:** Where is a patient in their disease trajectory?  
When will the disease progress? How will treatment affect  
disease progression?



Predicted risk of developing disease or predicting outcome



**Example:** Multiple myeloma

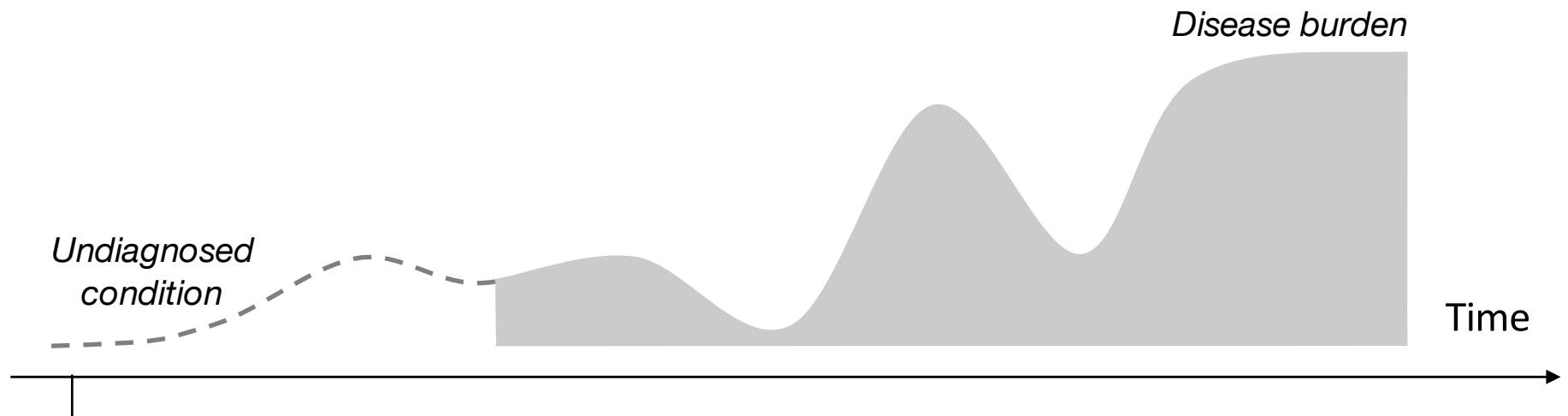
- ▶ Rare blood cancer
- ▶ MMRF CoMMpass Study has ~1000 patients

## Myeloma Staging Systems

| Stage | Durie-Salmon Staging System                                                                                                                                                                                                                                                                                                                                                                                   | Revised International Staging System                                                                                                                                                                                                     |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I     | <p>All of the following:</p> <ul style="list-style-type: none"> <li>○ Hemoglobin &gt;10.5 g/dL</li> <li>○ Serum calcium value normal or ≤12 mg/dL</li> <li>○ X-ray studies of bone, normal bone structure (scale 0) or solitary bone plasmacytoma only</li> <li>○ Low M-component production rate<br/>IgG value &lt;5 g/dL;<br/>IgA value &lt;3 g/dL</li> <li>○ Urine light chains &lt;4g/24 hours</li> </ul> | <ul style="list-style-type: none"> <li>○ Serum albumin &gt;3.5 g/dL</li> <li>○ Serum <math>\beta_2</math>-microglobulin &lt;3.5 mg/L</li> <li>○ No high-risk cytogenetics</li> <li>○ Normal serum lactate dehydrogenase level</li> </ul> |
| II    | <p>Neither stage I nor stage III</p> <ul style="list-style-type: none"> <li>○ A—No renal failure (creatinine ≤2 mg/dL)</li> <li>○ B—Renal failure (creatinine &gt;2 mg/dL)</li> </ul>                                                                                                                                                                                                                         | <p>Neither stage I nor stage III</p>                                                                                                                                                                                                     |
| III   | <ul style="list-style-type: none"> <li>○ Hemoglobin value &lt;8.5 g/dL</li> <li>○ Serum calcium value &gt;12 mg/dL</li> <li>○ X-ray studies of bone, &gt;3 lytic bone lesions</li> <li>○ High M-component production rate<br/>IgG value &gt;7 g/dL;<br/>IgA value &gt;5 g/dL</li> <li>○ Urine light chains &gt;12 g/24 hours</li> </ul>                                                                       | <ul style="list-style-type: none"> <li>○ Serum <math>\beta_2</math>-microglobulin &gt;5.5 mg/L</li> <li>○ High-risk cytogenetics<br/>t(4;14)<br/>t(14;16)<br/>del(17p)</li> <li>○ Elevated serum lactate dehydrogenase level</li> </ul>  |

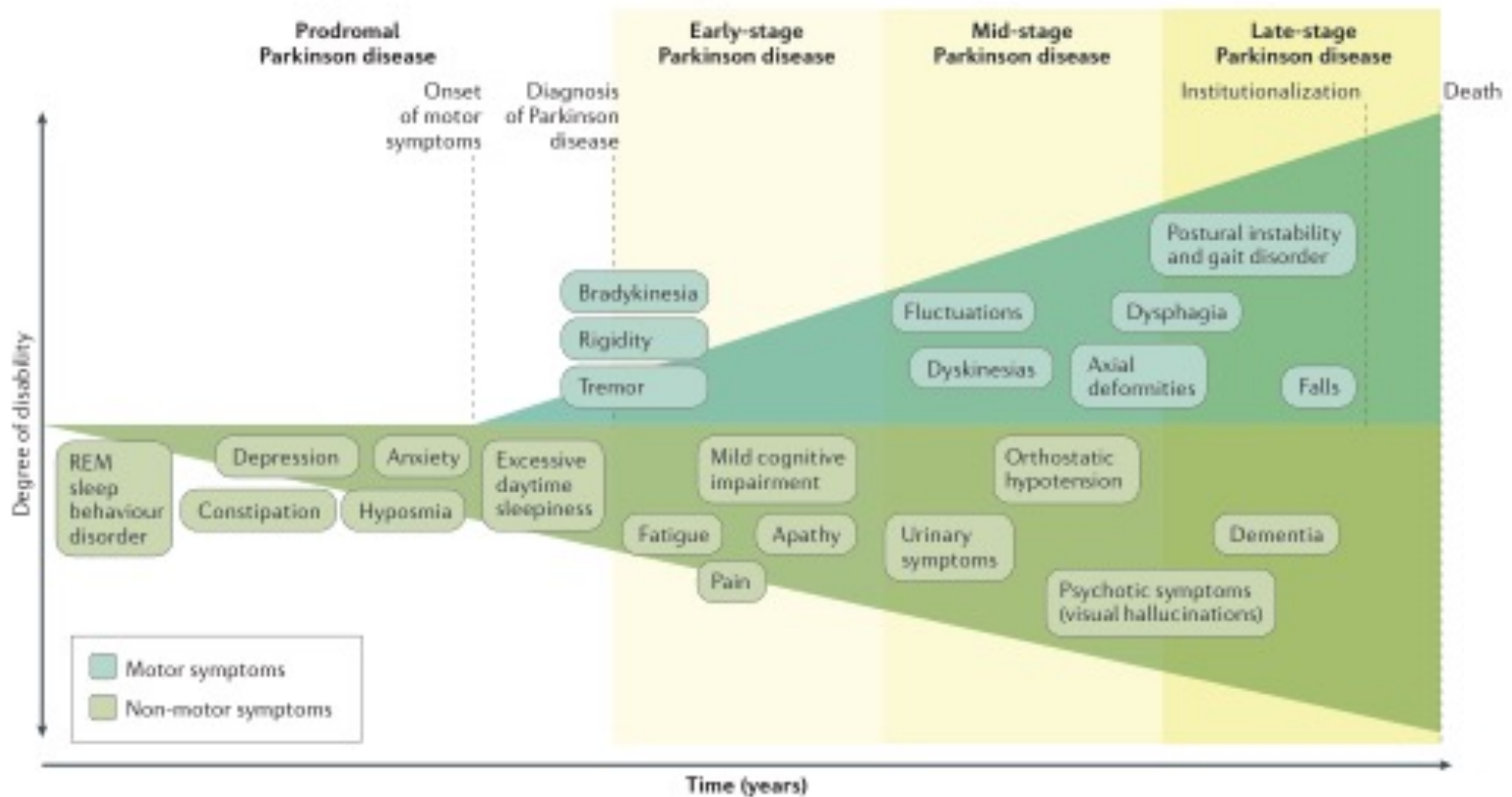
<https://www.lls.org/disease-information/myeloma/diagnosis/myeloma-staging>

## Descriptive: What does a typical trajectory look like?



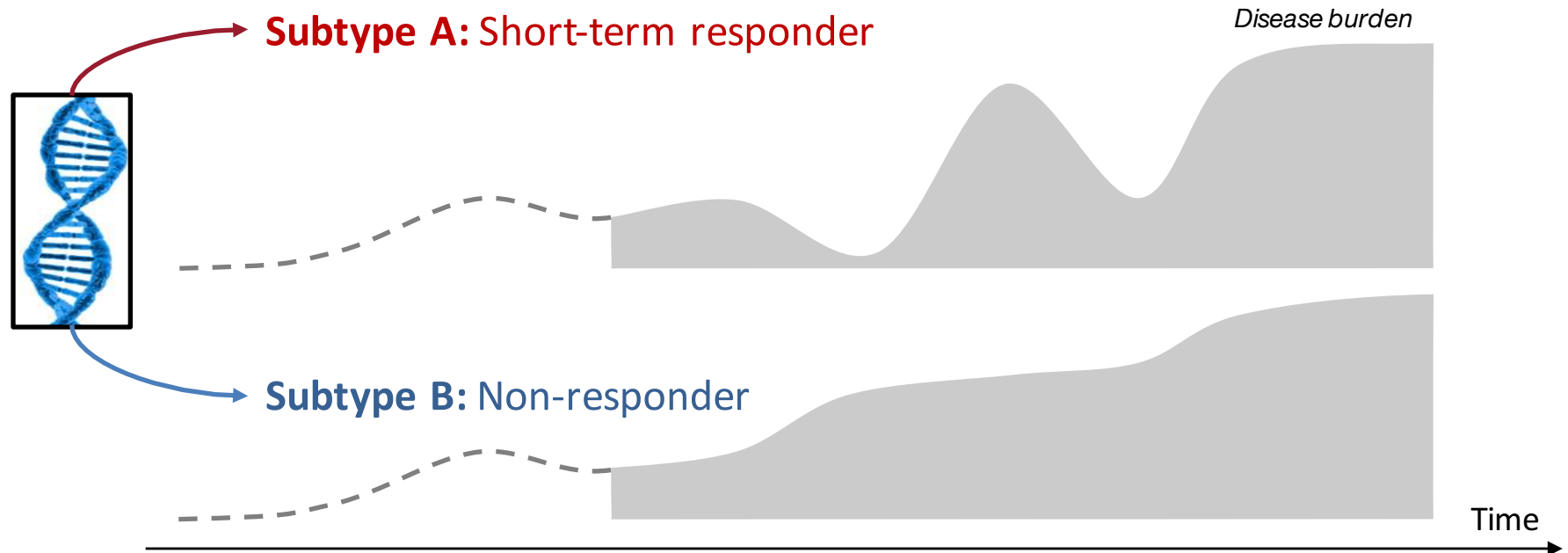
### Example: Parkinson's

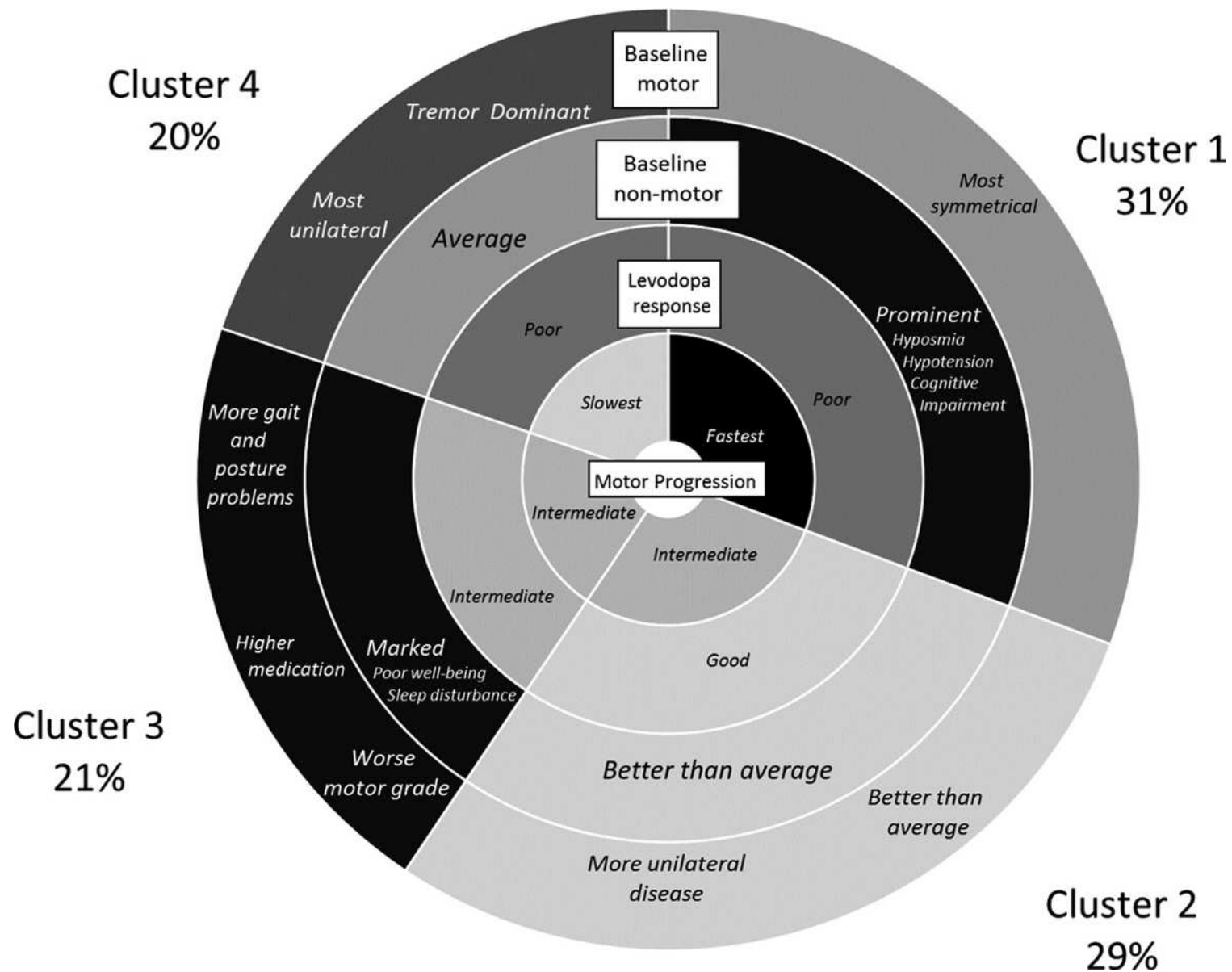
- ▶ Progressive nervous system disorder
- ▶ Affects 1 in 100 people over age 60
- ▶ PPMI dataset follows patients across time



[Poewe et al., Parkinson's disease. *Nature Reviews Disease Primers*, 2017]

# Subtyping: Can we re-define the disease altogether?

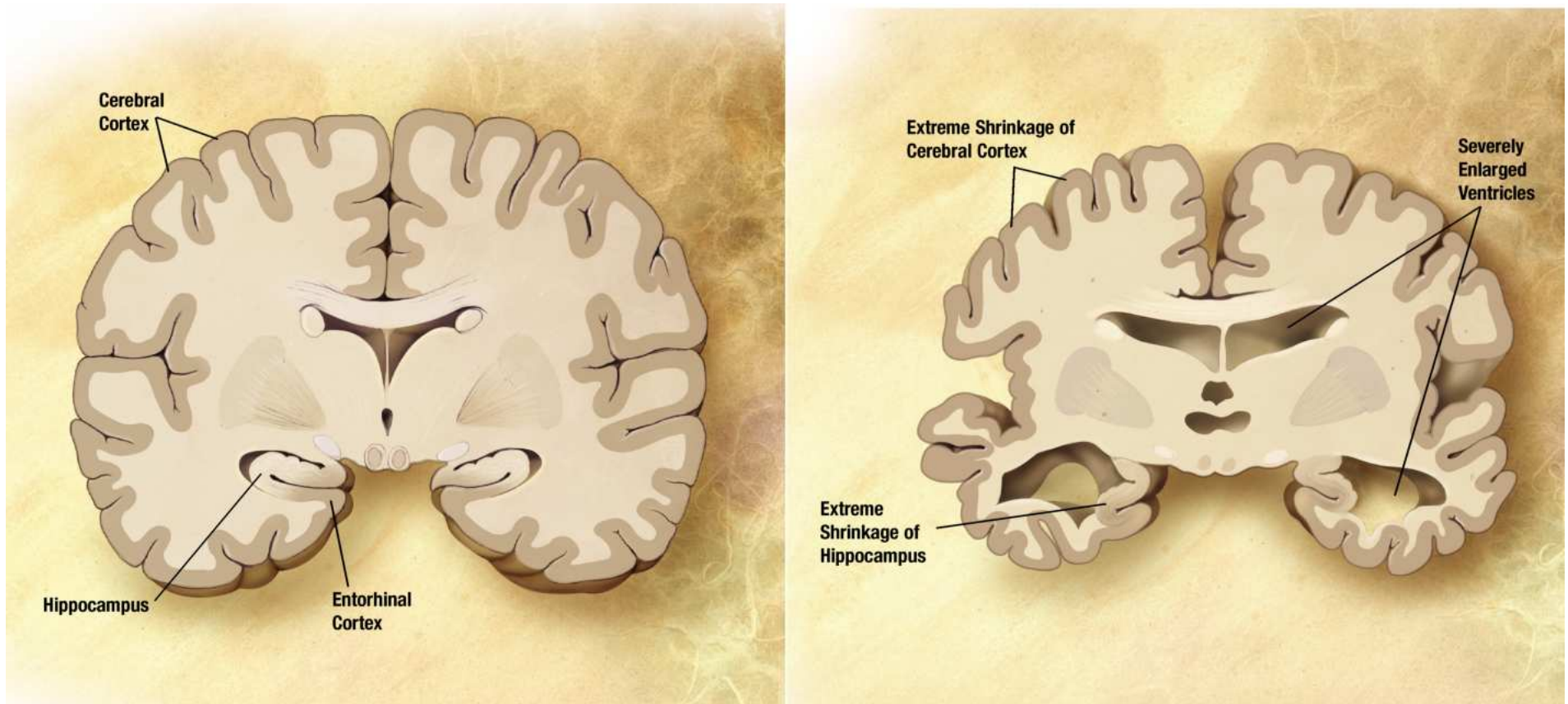




[Lawton et al., Developing and validating Parkinson's disease subtypes and their motor and cognitive progression. *J Neurol Neurosurg Psychiatry*, 2018]



# Predicting disease progression in Alzheimer's disease



[Image credit: Wikipedia; "Alzheimer's Disease Education and Referral Center, a service of the National Institute on Aging."]



Disease status  
quantified by  
cognitive score  
(continuous valued)

## MINI MENTAL STATE EXAMINATION (MMSE)

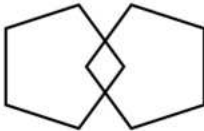
Name:

DOB:

Hospital Number:

One point for each answer

DATE:

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                          |                                                                          |                                                                          |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|--------------------------------------------------------------------------|--------------------------------------------------------------------------|
| <b>ORIENTATION</b><br>Year    Season    Month    Date    Time<br><br>Country    Town    District    Hospital    Ward/Floor                                                                                                                                                                                                                                                                                                                                                   | ...../ 5                                                                 | ...../ 5                                                                 | ...../ 5                                                                 |
| <b>REGISTRATION</b><br>Examiner names three objects (e.g. apple, table, penny) and asks the patient to repeat (1 point for each correct. THEN the patient learns the 3 names repeating until correct).                                                                                                                                                                                                                                                                       | ...../ 3                                                                 | ...../ 3                                                                 | ...../ 3                                                                 |
| <b>ATTENTION AND CALCULATION</b><br>Subtract 7 from 100, then repeat from result. Continue five times: 100, 93, 86, 79, 65. (Alternative: spell "WORLD" backwards: DLROW).                                                                                                                                                                                                                                                                                                   | ...../ 5                                                                 | ...../ 5                                                                 | ...../ 5                                                                 |
| <b>RECALL</b><br>Ask for the names of the three objects learned earlier.                                                                                                                                                                                                                                                                                                                                                                                                     | ...../ 3                                                                 | ...../ 3                                                                 | ...../ 3                                                                 |
| <b>LANGUAGE</b><br>Name two objects (e.g. pen, watch).<br><br>Repeat "No ifs, ands, or buts".<br><br>Give a three-stage command. Score 1 for each stage. (e.g. "Place index finger of right hand on your nose and then on your left ear").<br><br>Ask the patient to read and obey a written command on a piece of paper. The written instruction is: "Close your eyes".<br><br>Ask the patient to write a sentence. Score 1 if it is sensible and has a subject and a verb. | ...../ 2<br><br>...../ 1<br><br>...../ 3<br><br>...../ 1<br><br>...../ 1 | ...../ 2<br><br>...../ 1<br><br>...../ 3<br><br>...../ 1<br><br>...../ 1 | ...../ 2<br><br>...../ 1<br><br>...../ 3<br><br>...../ 1<br><br>...../ 1 |
| <b>COPYING:</b> Ask the patient to copy a pair of intersecting pentagons<br><br>                                                                                                                                                                                                                                                                                                         | ...../ 1                                                                 | ...../ 1                                                                 | ...../ 1                                                                 |
| <b>TOTAL:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                | ...../ 30                                                                | ...../ 30                                                                | ...../ 30                                                                |

### MMSE scoring

24-30: no cognitive impairment

18-23: mild cognitive impairment

0-17: severe cognitive impairment

# Predicting disease progression in Alzheimer's disease

- Goal: Predict disease status in 6, 12, 24, 36, and 48 months
- Five different regression tasks?
- Challenge: data sparsity
  - Total number of patients is small
  - Labels are noisy
  - Due to censoring, fewer patients at later time points

# Predicting disease progression in Alzheimer's disease

- Goal: Predict disease status in 6, 12, 24, 36, and 48 months
- Five different regression tasks?
- Challenge: data sparsity

Number of patients M months after baseline  
(Alzheimer's Disease Neuroimaging Initiative)

| M06 | M12 | M24 | M36 | M48 |
|-----|-----|-----|-----|-----|
| 648 | 642 | 569 | 389 | 87  |

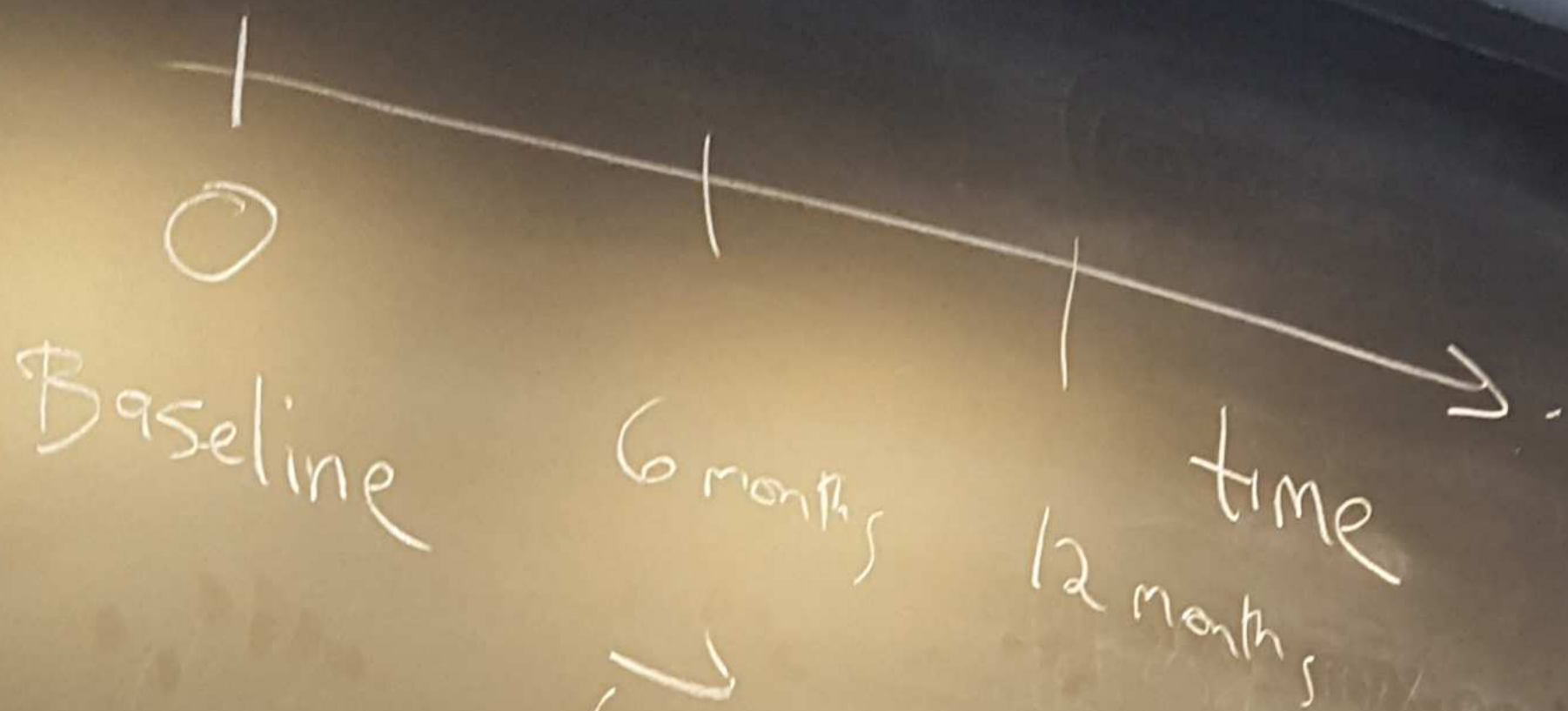
M06 = 6 months after baseline

[Zhou et al., KDD '12]

# Multi-task learning

- Goal: Predict disease status in 6, 12, 24, 36, and 48 months
- Rather than learn several independent models, view as *multi-task* learning
  - Select common set of biomarkers for all time points
  - Also allow for specific set of biomarkers at different time points
  - Incorporate temporal smoothness in models

[Zhou et al., KDD '12]



6 months  
12 months

$$D_1 = \{(\tilde{x}, y) \mid \tilde{w}, y \in \{0, 1\}\}$$

$$D_2 = \{(\tilde{x}, y) \mid \tilde{w}_2$$

$$k=1, 2$$

$$\min_{\{\tilde{w}_k\}_{k \in D_k}} \sum_k \sum_i \max(0, 1 - y_i \tilde{x}_i \cdot \tilde{w}_k) + \lambda \sum_k \|\tilde{w}_k - \tilde{w}_0\|_2^2$$

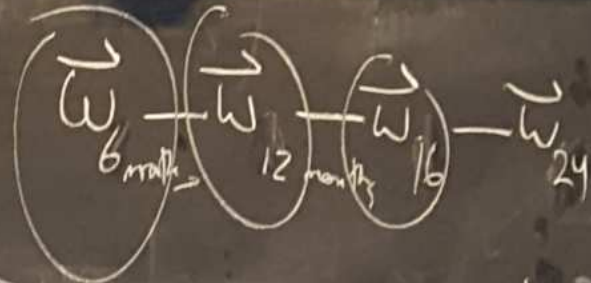


$\vec{w}_{16 \text{ months}}$

$\vec{w}_0$

$\vec{w}_{6 \text{ months}}$

$\vec{w}_{12 \text{ months}}$



$$\|\vec{w}_6 - \vec{w}_{12}\|_1 + \|\vec{w}_{12} - \vec{w}_{16}\|_1 + \|\vec{w}_{16} - \vec{w}_{24}\|_1$$

$$+ \frac{1}{n} \sum_{(i,j) \in E} \lambda_i \|\vec{w}_i - \vec{w}_j\|^2$$



# Convex fused sparse group lasso

- Simultaneously learn all 5 models by solving the following convex optimization problem:

$$\min_W L(W) + \lambda_1 \|W\|_1 + \lambda_2 \|RW^T\|_1 + \lambda_3 \|W\|_{2,1}$$

- Squared loss:  $L(W) = \|S \odot (XW - Y)\|_F^2$   
( $S$  is a mask to account for labels missing in subset of tasks)

- Group Lasso penalty  $\|W\|_{2,1}$  given by  $\sum_{i=1}^d \sqrt{\sum_{j=1}^t W_{ij}^2}$

- $R =$ 

5

|   |    |    |    |  |
|---|----|----|----|--|
| 1 | -1 |    |    |  |
|   | 1  | -1 |    |  |
|   |    | 1  | -1 |  |

4

[Zhou et al., KDD '12]

# Features

MRI scans (white matter parcellation volume, etc.) +

|                           |                                                                                                                       |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Demographic               | age, years of education, gender                                                                                       |
| Genetic                   | ApoE- $\epsilon$ 4 information                                                                                        |
| Baseline cognitive scores | MMSE, ADAS-Cog, ADAS-MOD, ADAS subscores, CDR, FAQ, GDS, Hachinski, Neuropsychological Battery, WMS-R Logical Memory  |
| Lab tests                 | RCT1, RCT11, RCT12, RCT13, RCT14, RCT1407, RCT1408, RCT183, RCT19, RCT20, RCT29, RCT3, RCT392, RCT4, RCT5, RCT6, RCT8 |

**371 in total**

[Zhou et al., KDD '12]

# Results (averaged over 5 time points)

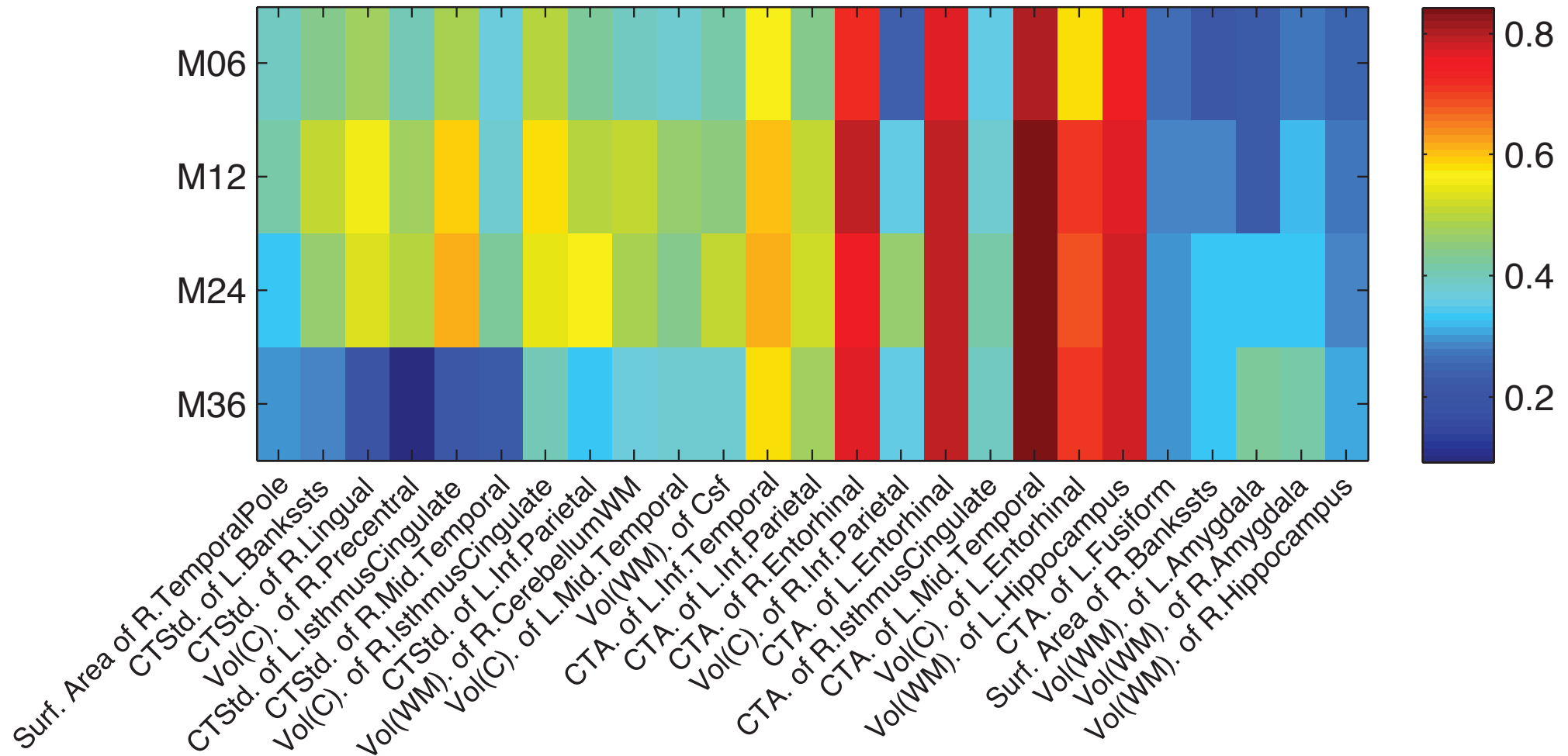
| Baseline – independent regressors |                   | Temporal smoothing helps! |                   |                                     |
|-----------------------------------|-------------------|---------------------------|-------------------|-------------------------------------|
|                                   |                   | $\lambda_2 = 20$          | $\lambda_2 = 50$  | $\lambda_2 = 100$                   |
|                                   | Ridge             | cFSGL1                    | cFSGL2            | cFSGL3                              |
| Target: MMSE                      |                   |                           |                   |                                     |
| nMSE                              | $0.548 \pm 0.057$ | $0.428 \pm 0.052$         | $0.400 \pm 0.053$ | <b><math>0.395 \pm 0.052</math></b> |
| R                                 | $0.689 \pm 0.030$ | $0.772 \pm 0.030$         | $0.790 \pm 0.032$ | <b><math>0.796 \pm 0.031</math></b> |

nMSE – normalized mean squared error. Smaller is better

R – average  $R^2$  (correlation coefficient). Larger is better

$$\min_W L(W) + \lambda_1 \|W\|_1 + \lambda_2 \|RW^T\|_1 + \lambda_3 \|W\|_{2,1}$$

# Feature importance varies by time



(a) Target: ADAS-Cog (25 stable features)

# Can we use an unsupervised approach?

- Twin goals:
  - **Discover disease subtypes:**

Want to describe heterogeneity in a way that can be easy to act on (i.e., interpretable)

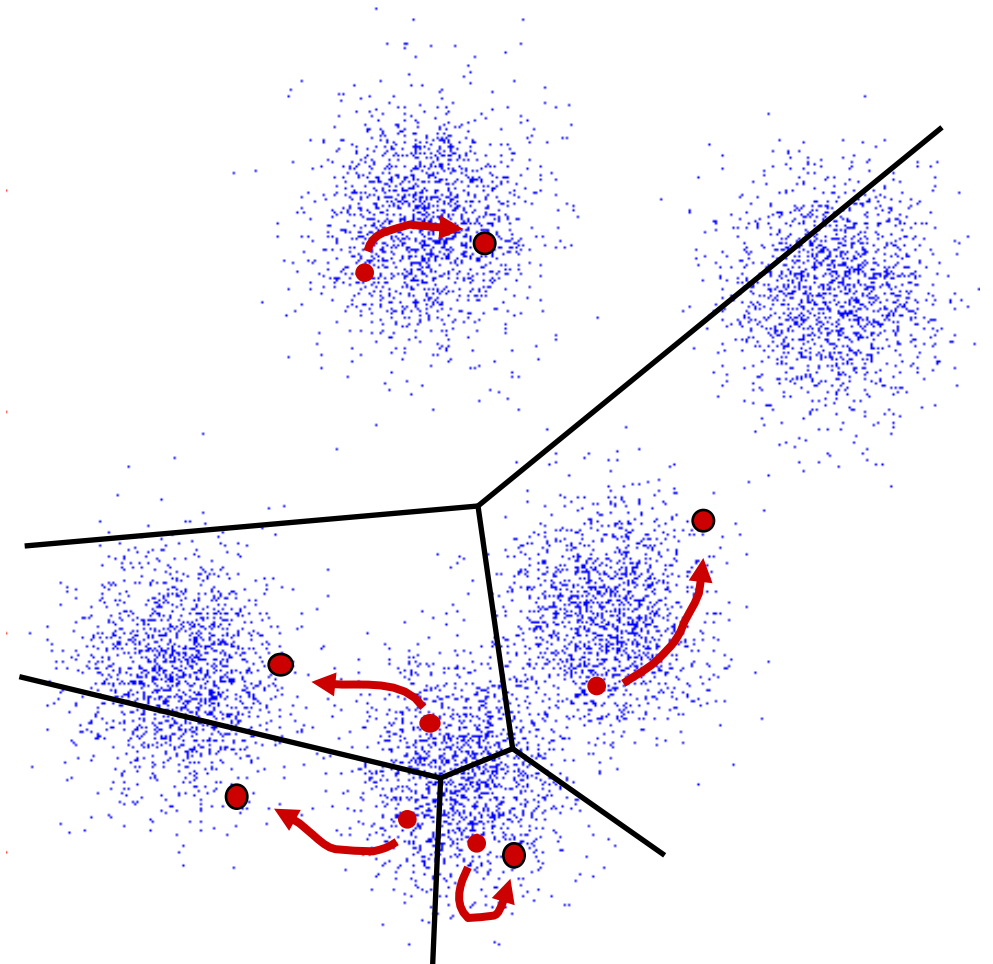
Not *just* interested in prediction – rather, identify cohorts for clinical trials, better understand disease mechanism

- **Make use of similarity of individuals at baseline**

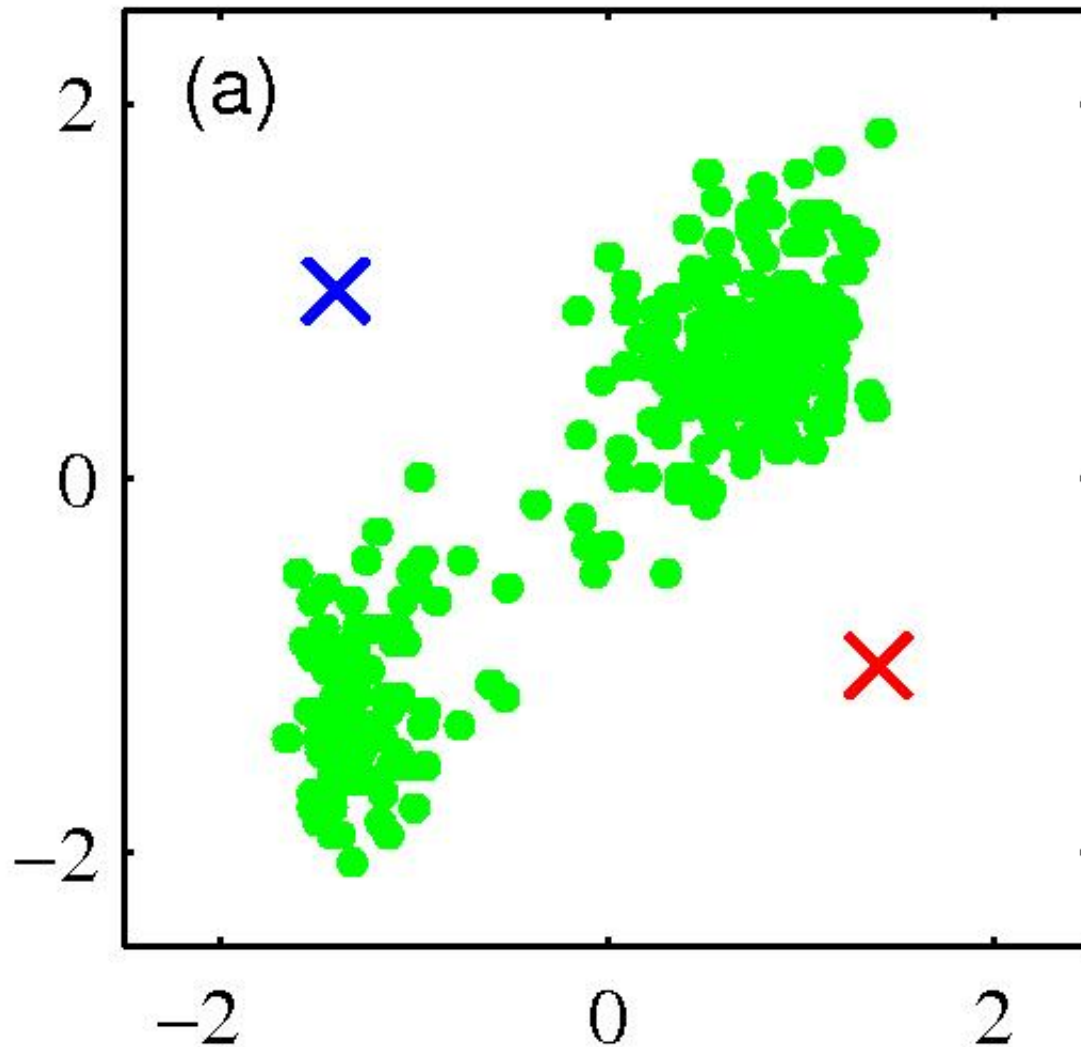
Dimensionality reduction to prevent overfitting

# K-Means

- An iterative clustering algorithm
  - **Initialize:** Pick  $K$  random points as cluster centers
  - **Alternate:**
    1. Assign data points to closest cluster center
    2. Change the cluster center to the average of its assigned points
  - **Stop** when no points' assignments change



# K-means clustering: Example

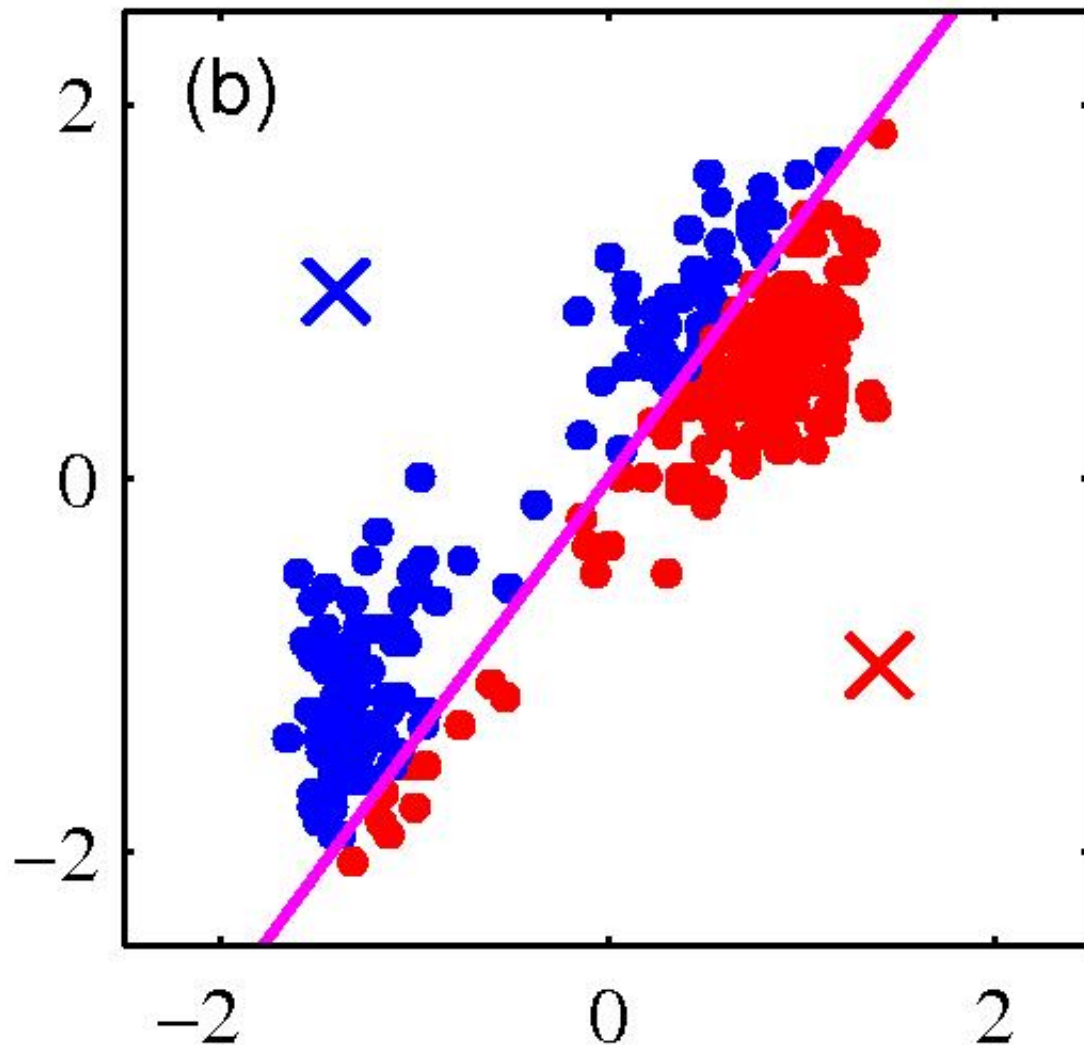


- Pick  $K$  random points as cluster centers (means)

Shown here for  $K=2$



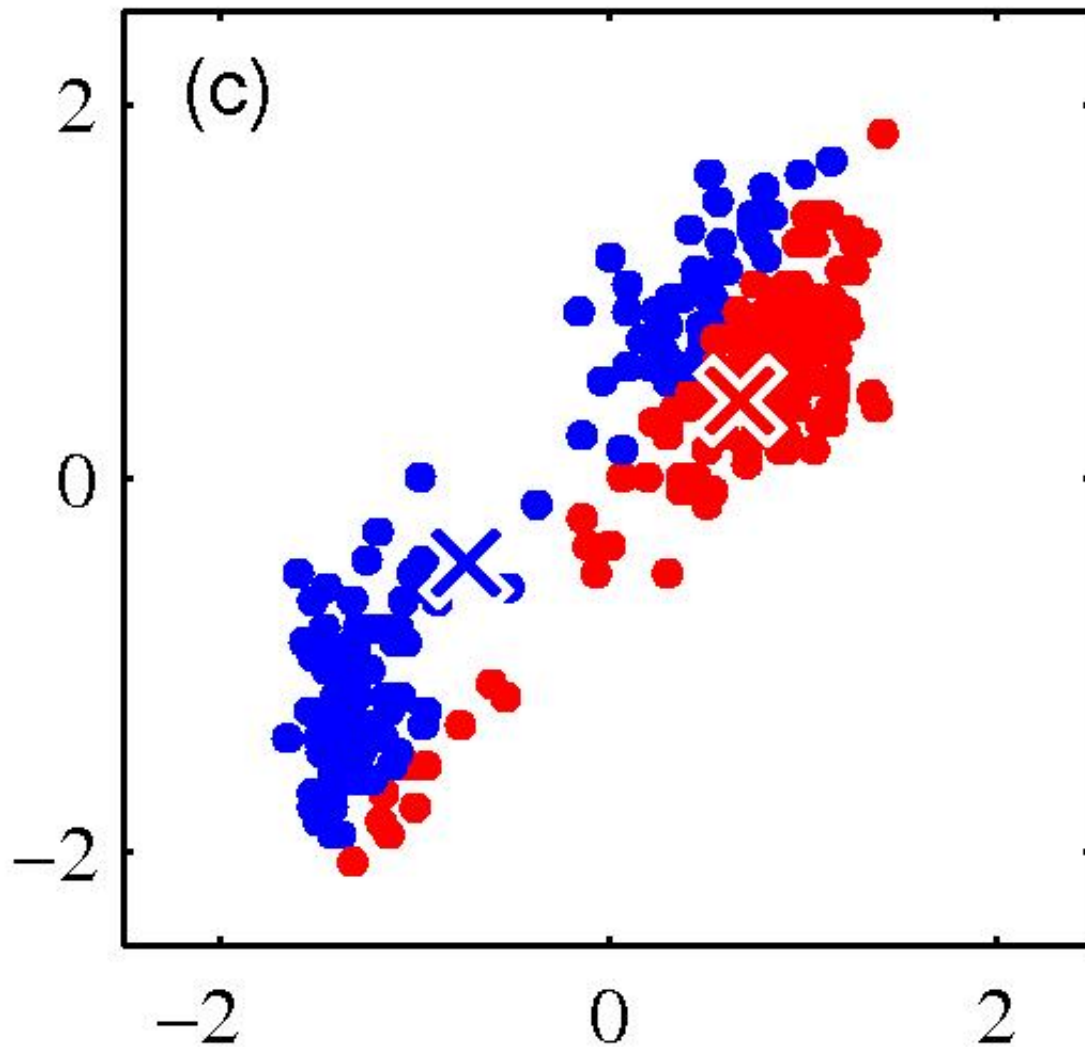
# K-means clustering: Example



Iterative Step 1

- Assign data points to closest cluster center

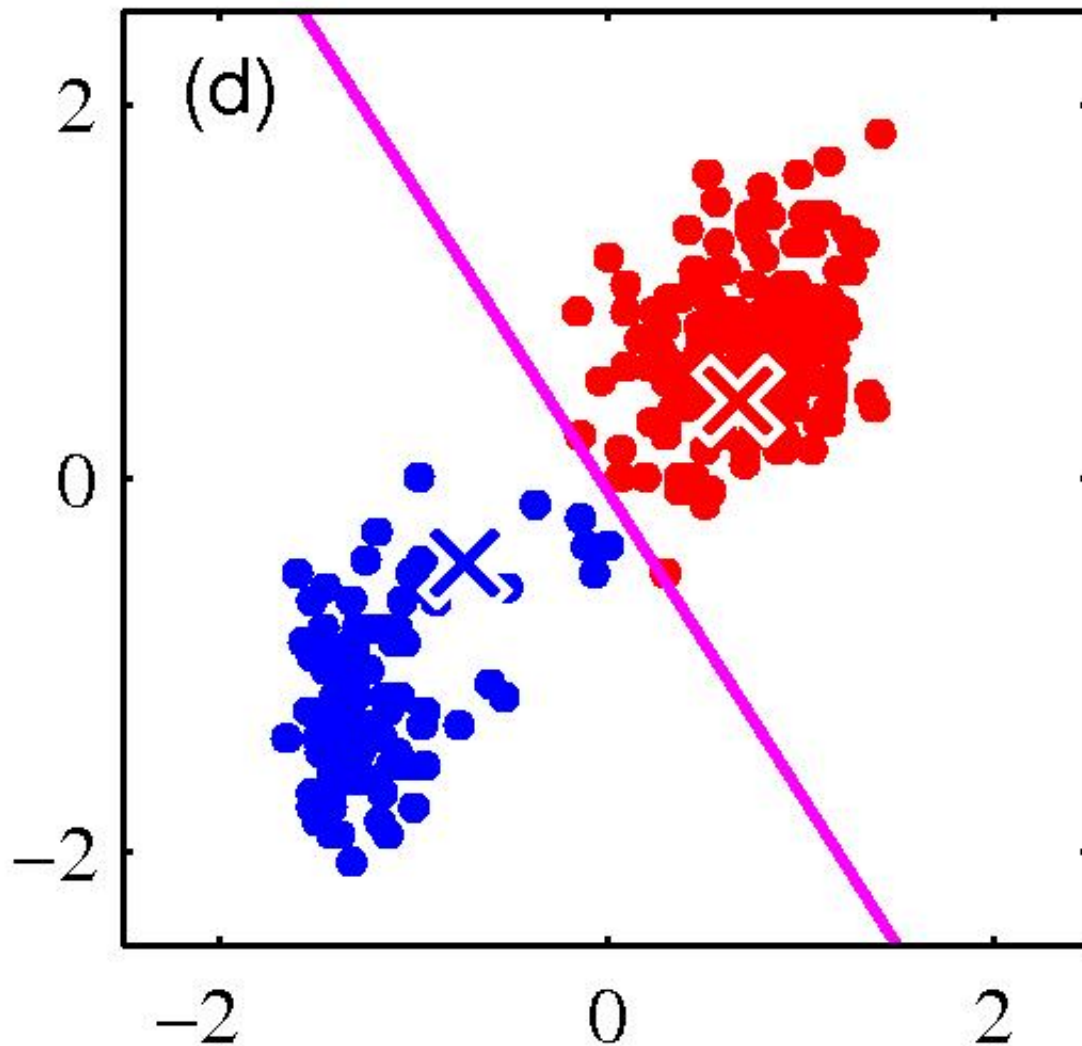
# K-means clustering: Example



## Iterative Step 2

- Change the cluster center to the average of the assigned points

# K-means clustering: Example



- Repeat until convergence

# Asthma: the problem

- 5 to 10% of people with severe asthma remain poorly controlled despite maximal inhaled therapy

[Holgate ST, Polosa R. The mechanisms, diagnosis, and management of severe asthma in adults. Lancet. 2006; 368:780–793]



[[whatasthmais.com](http://whatasthmais.com)]

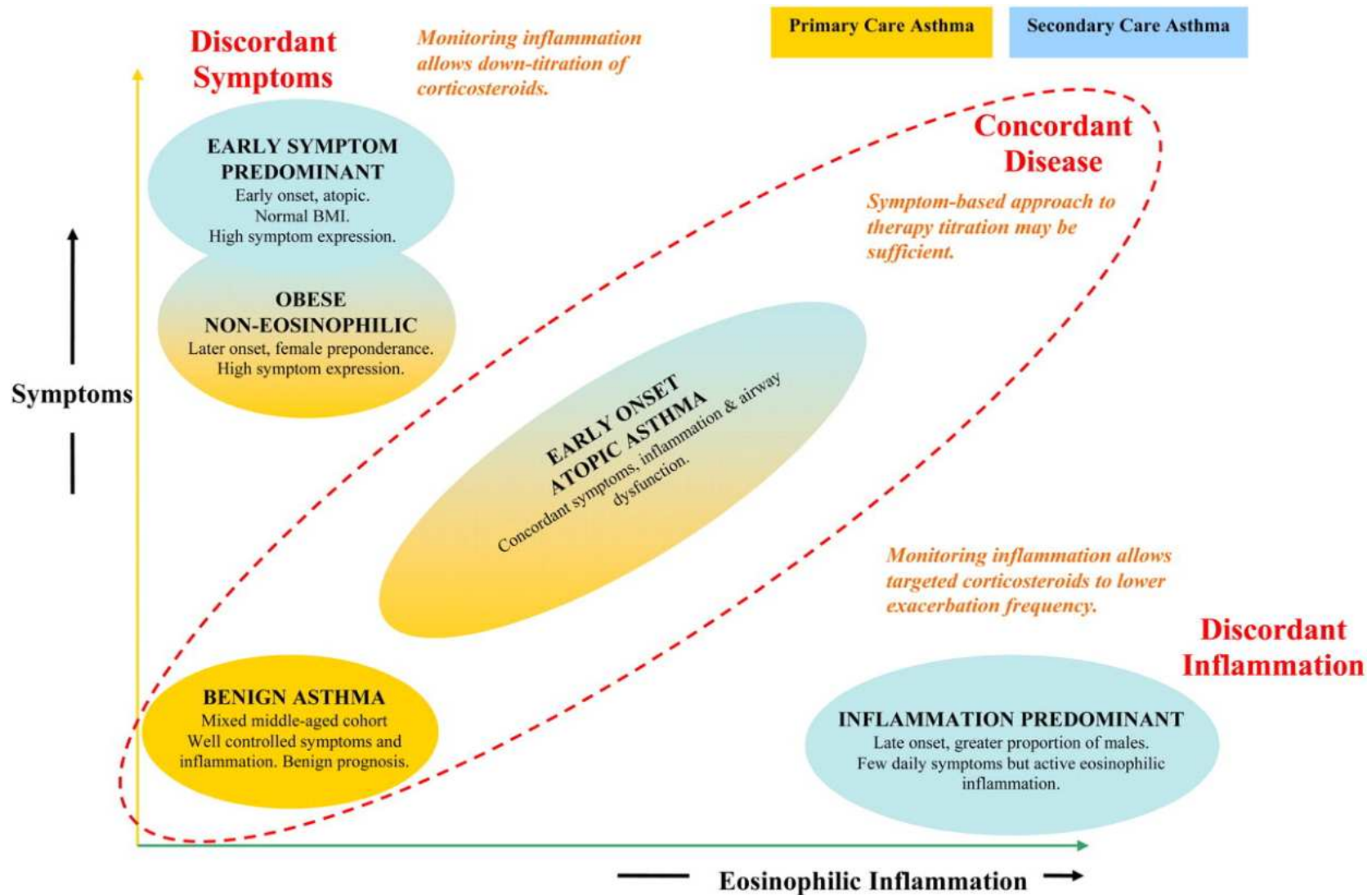
# Asthma: the question

**“It is now recognised that there are distinct asthma phenotypes and that distinct therapeutic approaches may only impinge on some aspects of the disease process within each subgroup”**

- What are the processes (genetic or environmental) that underlie different subtypes of asthma?
- Which aspects of airway remodelling are important in disease subtypes?
- What are the best biomarkers of disease progression or treatment response?
- Why are some patients less responsive to conventional therapies than others?

[Adcock et al., “New targets for drug development in asthma”. The Lancet, 2008]

# Discovering subtypes from data



[Haldar et al., *Am J Respir Crit Care Med*, 2008]

# The data

- All patients had physician diagnosis of asthma and at least one recent prescription for asthma therapy
- All were current nonsmokers
- *Data set #1*: 184 patients recruited from primary-care practices in the UK
- *Data set #2*: 187 patients from refractory asthma clinic in the UK
- *Data set #3*: 68 patients from 12 month clinical study
- Features: z scores for continuous variables, 0/1 for categorical
  - Some of the continuous variables log-transformed to approximate a normal distribution

[Haldar et al., *Am J Respir Crit Care Med*, 2008]

## Comparison of Baseline Characteristics in the three Asthma Populations

| Variable                                                     | Primary Care<br>(n = 184) | Secondary Care<br>(n = 187) | Longitudinal Cohort<br>(n = 68) |
|--------------------------------------------------------------|---------------------------|-----------------------------|---------------------------------|
| Sex, % female                                                | 54.4                      | 65.8                        | 47.1                            |
| Age, yr (SD)                                                 | 49.2 (13.9)               | 43.4 (15.9)                 | 52.4 (14.6)                     |
| Age of onset, yr (SD)                                        | 24.7 (19)                 | 20.3 (18.4)                 | 31.1 (23.7)                     |
| Atopic status, % positive                                    | 72.8                      | 73.8                        | 57.4                            |
| Body mass index, kg/m <sup>2</sup> (SD)                      | 27.5 (5.4)                | 28.5 (6.5)                  | 28.0 (5.9)                      |
| PC <sub>20</sub> methacholine <sup>†</sup> , mg/ml           | 1.04 (1.13)               | <sup>†</sup>                | 0.67 (0.68)                     |
| Peak flow variability, amp % mean                            | 17 (0.38)                 | 32.2 (0.48)                 | 13.8 (0.29)                     |
| FEV <sub>1</sub> change with bronchodilator, %               | 1.63 (1.16)               | 12.8 (0.41)                 | 3.2 (1.04)                      |
| Post-bronchodilator FEV <sub>1</sub> , % predicted           | 91.4 (21)                 | 82.1 (21.1)                 | 80.2 (20.6)                     |
| Sputum eosinophil count, %                                   | 1.32 (0.62)               | 2.9 (0.99)                  | 2.4 (0.81)                      |
| FE <sub>NO</sub> <sup>‡</sup> , ppb                          | 31.6 (0.33)               | 43 (0.32)                   | 4.32 (0.64) <sup>‡</sup>        |
| Sputum neutrophil count, %                                   | 55.09 (0.31)              | 46.7 (0.32)                 | 41.1 (0.35)                     |
| Modified JACS <sup>§</sup> (SD)                              | 1.36 (0.74)               | 2.02 (1.16)                 | 1.42 (1.26)                     |
| Dose of inhaled corticosteroid, BDP equivalent/ $\mu$ g (SD) | 632 (579)                 | 1,018 (539)                 | 1,821 (1,239)                   |
| Long-acting bronchodilator use, %                            | 40.2                      | 93                          | 86.7                            |

*Definition of abbreviations:* amp = amplitude; BDP = beclomethasone dipropionate; JACS = Juniper Asthma Control Score

[Haldar et al., *Am J Respir Crit Care Med*, 2008]



Clusters in  
primary  
care  
  
(found by  
K-means)

| Variable                                                                                       | Primary Care<br>(n = 184) | Cluster 1                                | Cluster 2                            | Cluster 3                 | Significance<br>(P Value)* |
|------------------------------------------------------------------------------------------------|---------------------------|------------------------------------------|--------------------------------------|---------------------------|----------------------------|
|                                                                                                |                           | Early-Onset<br>Atopic Asthma<br>(n = 61) | Obese<br>Noneosinophilic<br>(n = 27) | Benign Asthma<br>(n = 96) |                            |
| Sex <sup>†</sup> , % female                                                                    | 54.4                      | 45.9                                     | 81.5                                 | 52.1                      | 0.006                      |
| Age, yr (SD)                                                                                   | 49.2 (13.9)               | 44.5 (14.3)                              | 53.9 (14)                            | 50.8 (13)                 | 0.003                      |
| Age of onset <sup>†</sup> , yr (SD)                                                            | 24.7 (19)                 | 14.6 (15.4)                              | 35.3 (19.6)                          | 28.2 (18.3)               | <0.001                     |
| Atopic status <sup>†</sup> , % positive                                                        | 72.8                      | 95.1                                     | 51.9                                 | 64.6                      | <0.001                     |
| Body mass index <sup>†</sup> , kg/m <sup>2</sup> (SD)                                          | 27.5 (5.4)                | 26.1 (3.8)                               | 36.2 (5.5)                           | 26 (3.6)                  | <0.001                     |
| PC <sub>20</sub> methacholine <sup>†‡</sup> , mg/ml                                            | 1.04 (1.13)               | 0.12 (0.86)                              | 1.60 (0.93)                          | 6.39 (0.75)               | <0.001                     |
| PC <sub>20</sub> >8 mg/ml, n (%)                                                               | 64 (34.7)                 | 2 (3.3)                                  | 6 (22.2)                             | 56 (58.3)                 | <0.001                     |
| Peak flow variability <sup>†‡</sup> , amp % mean                                               | 17 (0.38)                 | 20 (0.47)                                | 21.9 (0.32)                          | 14.8 (0.32)               | 0.039                      |
| FEV <sub>1</sub> change with bronchodilator <sup>‡</sup> , %                                   | 1.63 (1.16)               | 4.5 (0.91)                               | 1.82 (1.16)                          | 0.83 (1.22)               | <0.001                     |
| Post-bronchodilator FEV <sub>1</sub> , % predicted                                             | 91.4 (21)                 | 86.9 (20.7)                              | 91.5 (21.4)                          | 94.2 (20.7)               | 0.107                      |
| Sputum eosinophil count <sup>†‡</sup> , %                                                      | 1.32 (0.62)               | 3.75 (0.64)                              | 1.55 (0.51)                          | 0.65 (0.44)               | <0.001                     |
| F <sub>ENO</sub> <sup>‡§</sup> , ppb                                                           | 31.6 (0.33)               | 57.5 (0.27)                              | 25.8 (0.29)                          | 22.8 (0.27)               | <0.001                     |
| Sputum neutrophil count <sup>‡</sup> , %                                                       | 55.09 (0.31)              | 45.87 (0.24)                             | 72.71 (0.13)                         | 57.56 (0.36)              | 0.038                      |
| Modified JACS <sup>†</sup> (SD)                                                                | 1.36 (0.74)               | 1.54 (0.58)                              | 2.06 (0.73)                          | 1.04 (0.66)               | <0.001                     |
| Dose of inhaled corticosteroid, BDP<br>equivalent/ $\mu$ g (SD)                                | 632 (579)                 | 548 (559)                                | 746 (611)                            | 653 (581)                 | 0.202                      |
| Long-acting bronchodilator use, %                                                              | 40.2                      | 34.4                                     | 48.2                                 | 41.7                      | 0.442                      |
| Previous hospital admission or emergency<br>attendance, no. per patient                        | 0.60 (1.57)               | 1.04                                     | 0.26                                 | 0.20                      | 0.037                      |
| Previous outpatient attendance, % attended                                                     | 15%                       | 22%                                      | 19%                                  | 6%                        | 0.121                      |
| Severe asthma exacerbations (requiring oral<br>corticosteroids) in past 12 mo, no. per patient | 1.25 (1.94)               | 1.86 (0.32)                              | 1.07 (0.32)                          | 0.39 (0.18)               | 0.002                      |

# Clusters in secondary care

| Variable                                                     | Secondary Care<br>(n = 187) | Cluster 1                                                                                                                                                                              | Cluster 2                             | Cluster 3                                | Cluster 4                               | Significance<br>(P Value)* |
|--------------------------------------------------------------|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|------------------------------------------|-----------------------------------------|----------------------------|
|                                                              |                             | Early Onset, Atopic<br>(n = 74)                                                                                                                                                        | Obese,<br>Noneosinophilic<br>(n = 23) | Early Symptom<br>Predominant<br>(n = 22) | Inflammation<br>Predominant<br>(n = 68) |                            |
| Sex <sup>†</sup> , % female                                  | 65.8                        | <div> <p>Resembled clusters from primary care – i.e., these are common across spectrum of severity</p> <p>Objective measures of disease severity show more advanced disease</p> </div> |                                       | 68.2                                     | 47.1                                    | <0.001                     |
| Age, yr (SD)                                                 | 43.4 (15.9)                 |                                                                                                                                                                                        |                                       | 35.5 (15.5)                              | 50.6 (15.1)                             | <0.001                     |
| Age of onset <sup>†</sup> , yr (SD)                          | 20.3 (18.4)                 |                                                                                                                                                                                        |                                       | 12.6 (15)                                | 32.6 (19.1)                             | <0.001                     |
| Atopic status <sup>†</sup> , % positive                      | 73.8                        |                                                                                                                                                                                        |                                       | 81.8                                     | 63.2                                    | 0.024                      |
| Body mass index <sup>†</sup> , kg/m <sup>2</sup> (SD)        | 28.5 (6.5)                  |                                                                                                                                                                                        |                                       | 23.6 (3.1)                               | 27 (3.9)                                | <0.001                     |
| Peak flow variability <sup>‡</sup> , amp % mean              | 32.2 (0.48)                 |                                                                                                                                                                                        |                                       | 24.2 (0.65)                              | 27.6 (0.36)                             | 0.002                      |
| FEV <sub>1</sub> change with bronchodilator <sup>‡</sup> , % | 12.8 (0.41)                 | 24.5 (0.31)                                                                                                                                                                            | 9.3 (0.35)                            | 4.5 (0.33)                               | 9.8 (0.34)                              | <0.001                     |
| Post-bronchodilator FEV <sub>1</sub> , % predicted (SD)      | 82.1 (21.1)                 | 79.0 (21.9)                                                                                                                                                                            | 79.0 (18.5)                           | 79.5 (26.1)                              | 87.2 (18.5)                             | 0.093                      |
| Sputum eosinophil count <sup>†‡</sup> , %                    | 2.9 (0.99)                  | 4.2 (0.76)                                                                                                                                                                             | 1.3 (1.01)                            | 0.1 (0.9)                                | 8.4 (0.64)                              | <0.001                     |
| FE <sub>NO</sub> <sup>‡§</sup> , ppb                         | 43 (0.32)                   | 51.2 (0.36)                                                                                                                                                                            | 24.2 (0.27)                           | 22.6 (0.30)                              | 53.1 (0.32)                             | <0.001                     |
| Sputum neutrophil count, % <sup>‡</sup>                      | 46.7 (0.32)                 | 45.4 (0.39)                                                                                                                                                                            | 49.3 (0.22)                           | 51.3 (0.23)                              | 45.9 (0.29)                             | 0.892                      |
| Modified JACS <sup>†</sup> (SD)                              | 2.02 (1.16)                 | 2.63 (0.93)                                                                                                                                                                            | 2.37 (1.09)                           | 2.11 (1.11)                              | 1.21 (0.95)                             | <0.001                     |
| Dose of inhaled corticosteroid, BDP equivalent/ $\mu$ g (SD) | 1,018 (539)                 | 1,168 (578)                                                                                                                                                                            | 1,045 (590)                           | 809 (396)                                | 914 (479)                               | 0.008                      |
| Long-acting bronchodilator use, %                            | 93.0                        | 91.9                                                                                                                                                                                   | 95.4                                  | 90.9                                     | 94.1                                    | 0.999                      |

# How should we treat asthma?

- Now we use 3<sup>rd</sup> dataset – 68 patients over 12 months
- Randomized control trial with two arms:
  - Standard clinical care (“clinical”)
  - Regular monitoring of airway inflammation using induced sputum, to titrate steroid therapy to maintain normal eosinophil counts (“sputum”)
- Original study found no difference in corticosteroid usage
  - But, this could have been explained by heterogeneity in treatment response!

# Patients in different clusters respond differently to treatment! (analysis using 3<sup>rd</sup> dataset from 12 month study)

| Cluster<br>(found using <i>baseline</i> data) | Outcomes                                                     | Treatment strategy           |                           | Significance |
|-----------------------------------------------|--------------------------------------------------------------|------------------------------|---------------------------|--------------|
|                                               |                                                              | Clinical<br>( <i>n</i> = 10) | Sputum<br>( <i>n</i> = 8) |              |
| 1: Obese female                               | Δ Inhaled corticosteroid dose <sup>*</sup> /μg per day (SEM) | −400 (328)                   | −462 (271)                | 0.89         |
|                                               | Severe exacerbation frequency over 12 mo (SEM)               | 1.40 (0.78)                  | 1.50 (0.80)               | 0.93         |
|                                               | Number commenced on oral corticosteroids                     | 2                            | 1                         | 0.59         |
|                                               |                                                              | Clinical ( <i>n</i> = 15)    | Sputum ( <i>n</i> = 24)   |              |
| 2: Inflammation predominant                   | Δ Inhaled corticosteroid dose <sup>*</sup> /μg per day (SEM) | +753 (334)                   | +241 (233)                | 0.22         |
|                                               | Severe exacerbation frequency over 12 mo (SEM)               | 3.53 (1.18)                  | 0.38 (0.13)               | 0.002        |
|                                               | Number commenced on oral corticosteroids                     | 2                            | 9                         | 0.17         |
|                                               |                                                              | Clinical ( <i>n</i> = 7)     | Sputum ( <i>n</i> = 4)    |              |
| 3: Early symptom predominant                  | Δ Inhaled corticosteroid dose <sup>*</sup> /μg per day (SEM) | +1,429 (429)                 | −400 (469)                | 0.022        |
|                                               | Severe exacerbation frequency over 12 mo (SEM)               | 5.43 (1.90)                  | 2.50 (0.87)               | 0.198        |
|                                               | Number commenced on oral corticosteroids                     | 6                            | 0                         | Undefined    |

[Haldar et al., *Am J Respir Crit Care Med*, 2008]

# Summary – two approaches

- **Supervised:**  
predict future disease status
- **Unsupervised:**  
which patients look similar / different? Do clusters have different outcomes?

# Limitations that we'll address in the next lecture

- Can't differentiate between *stage* and *subtype*
  - Patients assumed to be aligned at baseline
- Only make use of one time point per patient
- Assumes single factor (cluster) explains all variation
- Distance function is particularly simplistic
- Either supervised or unsupervised, but not both – how to combine?